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Theorem 0.1. A class constructed with conjugation of a formula and the uniqueness of object satisfying the same formula has at most one element.

$$(!w : w \in \{y|\varphi \wedge (!y : \varphi)\})$$

Proof.
Assume $w_1 \in \{y|\varphi \wedge (!y : \varphi)\}$ and $w_2 \in \{y|\varphi \wedge (!y : \varphi)\}$.
To show: $w_1 = w_2$.

$$w_1 \in \{y|\varphi \wedge (!y : \varphi)\} \implies (w_1 \text{ set}) \wedge \varphi[w_1/y] \wedge (!y : \varphi)$$

$$w_2 \in \{y|\varphi \wedge (!y : \varphi)\} \implies (w_2 \text{ set}) \wedge \varphi[w_2/y] \wedge (!y : \varphi)$$

$$(!y : \varphi) \ \& \ (w_1 \text{ set}) \wedge \varphi[w_1/y] \ \& \ (w_1 \text{ set}) \wedge \varphi[w_2/y] \implies w_1 = w_2$$

Done.

Theorem 0.2.

$$(A \subseteq B) \wedge (!x : x \in B) \longrightarrow (!x : x \in A)$$

Proof.
Assume $(A \subseteq B)$ and $(!x : x \in B)$.

$(!x : x \in B) \implies (\forall x_1 \forall x_2 : x_1 \in B \wedge x_2 \in B \longrightarrow x_1 = x_2)$ (by def of uniqueness quantifier **??**)

Claim $(\forall x_1 \forall x_2 : x_1 \in A \wedge x_2 \in A \longrightarrow x_1 = x_2)$.
Fix x_1 and x_2. Assume $x_1 \in A$ and $x_2 \in A$.

$$A \subseteq B \ \& \ x_1 \in A \implies x_1 \in B$$

$$A \subseteq B \ \& \ x_2 \in A \implies x_2 \in B$$

$$(!x : x \in B) \ \& \ x_1 \in B \ \& \ x_2 \in B \implies x_1 = x_2$$

Claim is proven.

$(\forall x_1 \forall x_2 : x_1 \in A \wedge x_2 \in A \longrightarrow x_1 = x_2) \implies (!x : x \in A)$ (by def of uniqueness quantifier **??** costructor)

Done.